

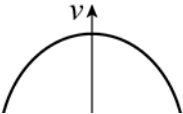
6	B
---	---

7	B
---	---

8	B
---	---

7	B
---	---

20	D	59 MPa	AO2
----	---	--------	-----

07	D	time period of rotation of m
		

Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Arrow drawn from water towards X ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	Idea that reaction force = 0 N	Treat consideration of the tension as neutral.	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	Evidence of weight = mass x centripetal acceleration ✓ (To give) evidence of $\omega = \sqrt{\frac{g}{r}}$ ✓ = 3.4(0) (rad s⁻¹) ✓	e.g. $mg = mr\omega^2$ For MP2 credit either manipulation or substitution Note: ans= 3.39723 using g = 9.81 ans=3.39550 using g = 9.8	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.4	Idea of reducing mass of water ✓ ...reference to (Max $T =) mg + mr\omega^2$ ✓ Idea of reducing radius of circle ✓ ...reference to $mr\omega^2$ ✓	If no reference to mg in the discussion, do not award MP2	4	AO3
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

Question	Answers	Additional comments/Guidelines	Mark	AO
07.1	Arrow from J towards the centre of the circular plate. The line of action of the arrow should pass through the centre of the coin and the centre of the plate (by eye) ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
07.2	Any two from: ✓✓ • $F = 0.85 \text{ mg}$ • use of $m\omega^2 r$ • $2\pi r = 0.94$ $7.5 \text{ (rad s}^{-1}\text{)} \checkmark$	$r = \frac{0.94}{2\pi} = 0.1496$ $\omega^2 = \frac{0.85g}{r}$ is awarded both mp1 and mp2 $\omega^2 = \frac{0.85g}{r} = \frac{0.85 \times 9.81}{0.1496} = 55.74$ $\omega = 7.47 \text{ rad s}^{-1}$	3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.3	$\omega_2 > \omega_1 \checkmark$ Idea that $F = m\omega^2 r$ and F and m are constant Or reference to $\omega^2 = \frac{0.85g}{r} \checkmark$	Allow $\frac{F}{m} = 0.85g$ for F and m are constant mp2 depends on mp1	2	AO2

Total			6	
--------------	--	--	----------	--

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

09	c	0.56 rad s^{-1}	AO2
----	---	---------------------------	-----

8	B
---	---